

Exact spectral-projection traces of classical unitary propagators

Candidate full-answer packet for arXiv:2302.00721

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Status: candidate full answer to the trace problem in Remark 3.5, likely valid, subject to human review. With the distribution-function convention used in the source, $U_t = e^{it\mathcal{L}}$ has $|U_t| = I$, so its spectral-projection trace is the two-step function $\tau(I)\mathbf{1}_{[0,1)}$. Thus the associated weak Lorentz norm is infinite when $\tau(I) = \infty$ and is independent of t and $\mathcal{L}$ when $\tau(I) < \infty$. The alternate literal normal-spectral-measure reading only counts the resonant eigenspace $e^{it\mathcal{L}} = 1$. Neither reading can encode classical dispersive decay.

1 Source problem and the notational issue

Gómez Cobos, Restrepo, and Ruzhansky study propagators affiliated with the right group von Neumann algebra $VN_R(G)$ of a locally compact separable unimodular group. In Remark 3.5 of arXiv:2302.00721v3 they observe that their fractional wave and Schrödinger estimates do not cover the classical endpoints and single out the following task [1]:

understand the behavior of $\tau(E_{(\lambda,+\infty)}(e^{it\mathcal{L}}))$ for the classical Schrödinger (wave) propagator; this remains open.

The source page is reproduced in Appendix A.

There is a small but decisive ambiguity. The source defines the distribution function of an affiliated operator T by

$$d_\lambda(T) = \tau(E_{(\lambda,+\infty)}(|T|)) , \quad \lambda \geq 0, \quad (1)$$

where $|T| = (T^*T)^{1/2}$. This is the projection relevant to generalized singular numbers and noncommutative Lorentz norms. Remark 3.5 omits the modulus, even though $e^{it\mathcal{L}}$ is unitary rather than positive. We compute both the intended distribution projection and the literal spectral projection of the normal operator. Both are elementary and both obstruct the suggested route to dispersion.

Idea of the proof

Unitary propagation changes phase but not size. Consequently the modulus of the propagator is exactly the identity, so every singular value equals one. This makes its distribution trace completely independent of time and of the generator. If one instead takes the spectral projection of the unitary operator itself, a positive real interval meets the unit circle only at the point 1; functional calculus then pulls this point back to the resonant levels $t\lambda \in 2\pi\mathbb{Z}$ of the generator. The first interpretation forgets all phase, while the second sees only exact resonances. Classical dispersion comes from cancellation between phases and therefore lies between these two phase-insensitive extremes.

2 Exact distribution function and Lorentz norm

Let $(\mathcal{M}, \tau)$ be a semifinite von Neumann algebra, and let $\mathcal{L}$ be a positive self-adjoint operator affiliated with $\mathcal{M}$. The bounded Borel functional calculus gives $U_t = e^{it\mathcal{L}} \in \mathcal{M}$ for every $t \in \mathbb{R}$. Write $T = \tau(I) \in (0, +\infty]$.

Theorem 1 (complete distribution computation). *For every $t \in \mathbb{R}$ and $\lambda \geq 0$,*

$$d_\lambda(U_t) = \tau(E_{(\lambda, +\infty)}(|U_t|)) = \begin{cases} T, & 0 \leq \lambda < 1, \\ 0, & \lambda \geq 1. \end{cases} \quad (2)$$

The generalized singular numbers are

$$\mu_s(U_t) = \begin{cases} 1, & 0 < s < T, \\ 0, & s \geq T, \end{cases}$$

with the second line absent when $T = +\infty$. Hence, for every $1 \leq r < \infty$,

$$\|U_t\|_{L^{r,\infty}(\mathcal{M})} = T^{1/r}, \quad (3)$$

where $+\infty^{1/r} = +\infty$.

Proof. The spectral theorem gives $U_t^* = e^{-it\mathcal{L}}$ and therefore

$$U_t^* U_t = e^{-it\mathcal{L}} e^{it\mathcal{L}} = I.$$

Thus $|U_t| = (U_t^* U_t)^{1/2} = I$. The spectral measure of I is supported at 1, so $E_{(\lambda, +\infty)}(I)$ equals I for $0 \leq \lambda < 1$ and equals 0 for $\lambda \geq 1$. Taking the trace proves (2).

By definition, $\mu_s(U_t) = \inf\{\lambda \geq 0 : d_\lambda(U_t) \leq s\}$. Formula (2) gives the displayed singular-number formula. Finally,

$$\|U_t\|_{L^{r,\infty}(\mathcal{M})} = \sup_{s>0} s^{1/r} \mu_s(U_t) = T^{1/r},$$

with the supremum approached from below when $T < \infty$. □

Corollary 1 (sharp obstruction to the proposed route). *If $\tau(I) = +\infty$, then $e^{it\mathcal{L}} \notin L^{r,\infty}(\mathcal{M})$ for every finite r . If $\tau(I) < +\infty$, its weak Lorentz norm is the constant $\tau(I)^{1/r}$, independent of t and $\mathcal{L}$. The same conclusions hold for the half-wave unitary $e^{it\sqrt{\mathcal{L}}}$.*

Proof. Apply Theorem 1 first to $\mathcal{L}$ and then to the positive self-adjoint affiliated operator $\sqrt{\mathcal{L}}$. □

3 The alternate literal spectral-projection reading

Let F_t denote the projection-valued spectral measure of the normal operator U_t . Now regard $(\lambda, +\infty)$ as a subset of the real axis inside $\mathbb{C}$.

Theorem 2 (literal normal spectral projection). *Suppose $t \neq 0$. For $0 \leq \lambda < 1$,*

$$F_t((\lambda, +\infty)) = F_t(\{1\}) = E^{\mathcal{L}}(\{s \geq 0 : ts \in 2\pi\mathbb{Z}\}), \quad (4)$$

where $E^{\mathcal{L}}$ is the spectral measure of $\mathcal{L}$. For $\lambda \geq 1$ the projection is zero. At $t = 0$, it is I below threshold 1 and zero at or above 1.

Proof. The spectrum of U_t lies in the unit circle. For $0 \leq \lambda < 1$, the real interval $(\lambda, +\infty)$ meets that circle only at 1; for $\lambda \geq 1$ the intersection is empty. Since the spectral measure of $e^{it\mathcal{L}}$ is the pushforward of $E^\mathcal{L}$ under $s \mapsto e^{its}$, the preimage of $\{1\}$ is exactly the set in (4). The case $t = 0$ is immediate from $U_0 = I$. $\square$

This literal quantity is well-defined, but it is not the distribution function (1) used by the source's Lorentz-space method. It measures only atomic spectral mass at exact resonances and has no general relation to dispersive cancellation.

4 Euclidean stress test: infinity or zero, but dispersion survives

Take $G = \mathbb{R}^n$ and $\mathcal{L} = -\Delta$. Under Fourier transform, $VN_R(\mathbb{R}^n)$ is the multiplication algebra on frequency space, $\mathcal{L}$ has symbol $|\xi|^2$, and the Plancherel trace is Lebesgue measure (up to the harmless Fourier normalization). Consequently $\tau(I) = +\infty$. Theorem 1 gives

$$\tau(E_{(\lambda, +\infty)}(|e^{it\mathcal{L}}|)) = +\infty \quad (0 \leq \lambda < 1),$$

and the weak L^r multiplier norm is infinite for finite r .

For the literal reading and $t \neq 0$, the resonant frequency set is

$$\{\xi \in \mathbb{R}^n : t|\xi|^2 \in 2\pi\mathbb{Z}\},$$

a countable union of spheres (and possibly the origin), hence a Lebesgue-null set. Its trace is zero. Thus the two readings produce opposite degenerate answers: infinity and zero.

Nevertheless, direct Fourier inversion gives the classical Schrödinger kernel, whose absolute value is $(4\pi|t|)^{-n/2}$ for $t \neq 0$, and hence

$$\|e^{it(-\Delta)}f\|_\infty \leq (4\pi|t|)^{-n/2}\|f\|_1$$

with the corresponding choice of Fourier normalization. This does not contradict Theorem 1: the kernel estimate exploits oscillatory cancellation, whereas generalized singular numbers take the modulus of the frequency multiplier and discard phase. It decisively shows that the proposed trace is not merely difficult to estimate; it is the wrong invariant for classical dispersion.

5 Scope, novelty check, and verification

Theorems 1 and 2 completely determine both natural meanings of the trace expression in Remark 3.5. They answer the stated trace problem and rule out using that quantity alone to extend the source's Lorentz-multiplier proof to the classical unitary endpoints. They do *not* classify L^p – L^q estimates for classical propagators on all locally compact groups; such estimates require additional structure capable of retaining phase or kernel cancellation.

The proof was checked directly against Definition 2.4 and Remark 3.5 of the current arXiv v3 source. The Euclidean model independently stress-tests both interpretations. The run registry and the solution, attempt, and proof-gap indexes were searched for the arXiv id and central trace/unitary-propagator phrases. Bounded exact-title, exact-phrase, arXiv-id, and close-variant web searches through 11 August 2026 found the source (last revised 16 February 2026, still stating the problem as open) and related spectral-multiplier work, but no later answer making this computation. This is not an exhaustive priority claim.

Human-review recommendation. Verify first that the intended projection in Remark 3.5 is the distribution projection from Definition 2.4; then $|e^{it\mathcal{L}}| = I$ makes the main conclusion immediate. If the omission of the modulus was deliberate, verify the pushforward identity (4). The principal review judgment is whether to label this a full answer to the stated trace problem or an obstruction to the broader, unstated goal of obtaining all classical dispersive estimates.

A Source-page evidence

References

- [1] Santiago Gómez Cobos, Joel E. Restrepo, and Michael Ruzhansky, *Boundedness of spectral multipliers on locally compact groups and applications*, arXiv:2302.00721v3, 2026.

Remark 3.5. We emphasize that Theorems 3.3, 3.4 do not cover the classical $\mathcal{L}$ -wave equation ($\beta = 2$) and $\mathcal{L}$ -Schrödinger equation ($\beta = 1$), respectively. For this case one would need a more delicate analysis of the problem, especially focusing on understanding the behavior of the trace of the spectral projection of the Schrödinger (wave) propagator $\tau(E_{(\lambda, +\infty)}(e^{it\mathcal{L}}))$. The latter remains an open problem.

4. EXAMPLES

In this section we provide some explicit examples to clarify and compare the wide class of groups and operators we can be treated with our results. We do it for the case of heat and wave type equations, but obviously the same examples work for the Schrödinger type equations.

Let us consider the following general $\mathfrak{D}$ -heat-wave type equation:

$$\begin{aligned} {}^C\partial_t^\alpha w(t, x) - \mathfrak{D}w(t, x) &= 0, \quad t > 0, \quad x \in G, \\ w(t, x)|_{t=0} &= w_0(x), \quad w_0 \in L^p(G), \\ \partial_t w(t, x)|_{t=0} &= w_1(x), \quad w_1 \in L^p(G), \end{aligned} \quad (4.1)$$

where $0 < \alpha < 2$ ($\alpha \neq 1$), $1 < p \leq 2$, $\mathfrak{D}$ is any positive linear left invariant operator on a locally compact separable unimodular group G . We use the first initial condition w_0 when $0 < \alpha < 1$. We consider both initial conditions w_0, w_1 when $1 < \alpha < 2$.

By Theorem 3.3, we get for any $1 < p \leq 2 \leq q < +\infty$ the following time decay rate for the solution of equation (4.1):

$$\|w(t, \cdot)\|_{L^q(G)} \leq C_{\alpha, \nu, Q, p, q} t^{-\alpha\lambda(\frac{1}{p} - \frac{1}{q})} (\|w_0\|_{L^p(G)} + t\|w_1\|_{L^p(G)}), \quad (4.2)$$

whenever

$$\tau(E_{(0, s)}(\mathfrak{D})) \lesssim s^\lambda, \quad s \rightarrow +\infty, \quad \text{for some } \lambda > 0, \quad \frac{1}{\lambda} \geq \frac{1}{p} - \frac{1}{q}.$$

Below we give several examples of equation (4.1) in different *Lie* groups and particular operators acting there, whose trace of the spectral projections behave like s^λ as $s \rightarrow +\infty$. Moreover, in some cases, we can claim the sharpness of the time-decay rate.

#	Group	Operator $\mathfrak{D}$	$\tau(E_{(0, s)}(\mathfrak{D}))$	Time Decay
1	The Euclidean space $\mathbb{R}^n$	$\Delta_{\mathbb{R}^n}$	$s^{n/2}$	$t^{-\alpha n/2(\frac{1}{p} - \frac{1}{q})}$
2	Any compact Lie group G	Δ_{sub}	$s^{Q/2}$	$t^{-\alpha Q/2(\frac{1}{p} - \frac{1}{q})}$
3	The Heisenberg group $\mathbb{H}^n$	$\mathcal{L}$	s^{n+1}	$t^{-\alpha(n+1)(\frac{1}{p} - \frac{1}{q})}$
4	Any graded Lie group G	$\mathcal{R}$	$s^{Q/\nu}$	$t^{-\alpha Q/\nu(\frac{1}{p} - \frac{1}{q})}$
5	Engel group $\mathfrak{B}_4$	$\mathfrak{D}_1$	s^3	$t^{-3\alpha(\frac{1}{p} - \frac{1}{q})}$
6	Cartan group $\mathfrak{B}_5$	$\mathfrak{D}_2$	$s^{9/2}$	$t^{-9\alpha/2(\frac{1}{p} - \frac{1}{q})}$
7	Any connected Lie group G	$\mathcal{L}$	$s^{Q_*/m}$	$t^{-\alpha Q_*/m(\frac{1}{p} - \frac{1}{q})}$

1. The Laplacian $\Delta_{\mathbb{R}^n}$ on the Euclidean space $\mathbb{R}^n$. By [2, Example 7.3], it follows that

$$\tau(E_{(0, s)}(\Delta_{\mathbb{R}^n})) \lesssim s^{n/2}, \quad s \rightarrow +\infty.$$

Figure 1: Page 14 of a local compilation of the official arXiv:2302.00721v3 source. Remark 3.5 states the classical propagator trace problem.